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Wireline release head

Abstract

A wireline release head system includes a housing, a cutter box situated within the housing, a fire control switch, and a wireline extending through the cutter box and in electrical communication with a control unit and the fire control switch. The cutter box includes a piston assembly operable to move within a first track and a cutter knife operable to move within a second track. The second track is angled relative to the first track. The fire control switch is in communication with a fire detonator, which is selectively activatable to detonate a power charge. An electrical signal selectively sent through the wireline from the control unit to the fire control switch detonates the power charge, which causes the piston assembly to move within the first track and contact the cutter knife causing the cutter knife to move within the second track, thereby severing the wireline.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/658,629, filed Apr. 8, 2022, and which is a continuation-in-part of U.S. patent application Ser. No. 17/228,201, filed on Apr. 12, 2021, and which is a continuation of U.S. patent application Ser. No. 17/062,925, filed on Oct. 5, 2020, and which claims priority to U.S. Provisional Patent Application No. 62/911,490 filed on Oct. 7, 2019. The entire contents of each of the above-referenced applications is hereby expressly incorporated herein by reference.

FIELD OF THE INVENTION

(1) The invention relates to cutting a wireline utilized in downhole well operations. More particularly, the invention includes a tool, system, and method for selectively disconnecting by

cutting a wireline above the cable connection point.

SUMMARY OF THE INVENTION

(2) The following represents a simplified summary of the disclosure in order to provide a basic understanding of some aspects of the disclosure. This summary is not an extensive overview of the disclosure. It is not intended to identify critical elements of the disclosure or to delineate the scope of the disclosure. Its sole purpose is to present some concepts of the disclosure in a simplified form as a prelude to the more detailed description that is presented elsewhere.

(3) According to one embodiment, a wireline release head system includes a housing, a cutter box situated within the housing, a fire control switch, and a wireline extending through the cutter box and being in electrical communication with a control unit and the fire control switch. The cutter box includes a piston assembly operable to move within a first track, and a cutter knife operable to move within a second track. The second track is angled relative to the first track. The fire control switch is in communication with a fire detonator, and the fire detonator is selectively activatable to detonate a power charge. An electrical signal selectively sent through the wireline from the control unit to the fire control switch detonates the power charge, which causes the piston assembly to move within the first track and contact the cutter knife. The contact with the cutter knife causes the cutter knife to move within the second track, and the movement of the cutter knife within the second track severs the wireline.

(4) In another embodiment, a wireline release head includes a cutter box having a piston assembly operable to move within a first track defined in the cutter box; a cutter knife operable to move within a second track defined in the cutter box; and a wireline receiving area for receiving at least a portion of a wireline. The second track is angled relative to the first track. Activation of movement of the piston assembly within the first track causes the piston assembly to contact the cutter knife, thereby causing the cutter knife to move within the second track. The movement of the cutter knife within the second track severs the wireline.

(5) According to still another embodiment, a method of severing a wireline includes first providing a wireline release head. The wireline release head includes a cutter box comprising a piston assembly operable to move within a first track, and a cutter knife operable to move within a second track; a fire control switch for selectively detonating a power charge; and a wireline extending through the cutter box and being in electrical communication with a control unit and the fire control switch. The method continues by selectively activating the control unit to send an electrical signal through the wireline to the fire control switch. The fire control switch detonates the power charge upon receipt of the electrical signal, which causes the piston assembly to move within the first track and contact the cutter knife. The contact causes the cutter knife to move within the second track, which severs the wireline.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a wireline tool string according to embodiments of the invention.
- (2) FIG. 2 is a cross section illustration of an embodiment of the invention in a run in hole position.
- (3) FIG. 2A is a top view of the invention taken along Line A-A in FIG. 2.
- (4) FIG. 2B is a top view of the invention taken along Line B-B in FIG. 2.
- (5) FIG. 2C is a top view of the invention taken along Line C-C in FIG. 2.
- (6) FIG. 3 is a cross section illustration of embodiment of the invention in a released position.
- (7) FIG. 3A is a top view of the invention taken along Line A-A in FIG. 3.
- (8) FIG. 3B is a top view of the invention taken along Line B-B in FIG. 3.
- (9) FIG. 3C is a top view of the invention taken along Line C-C in FIG. 3.

DETAILED DESCRIPTION

(10) Wireline operations are carried out in oil and gas wells to convey tools downhole into the well. A wide variety of downhole tools may be supported on a wireline, including tools to perform logging, perforating, and setting and retrieving operations. The tools typically include a combination of different tubular members threaded together to form a working unit, which is manipulated and controlled from the surface via the wireline. Although tools may be conveyed downhole on a tubing string that can withstand substantially higher extraction forces than a wireline, oftentimes a wireline is preferred because it saves substantial rig time in conveying tools downhole and positioning them within the well.

(11) A cable head, which connects the tools to the wireline, is typically provided with a release mechanism to permit the wireline to be disconnected from the tools in the event the tools become stuck downhole. For normal wireline operations performed in oil and gas wells, it is beneficial to have a primary and contingency means to disconnect the wireline from the tools attached at the end of the wireline if the tool assembly gets stuck to retrieve the wire and more easily retrieve the stuck tool assembly to resume normal activity. Typically, a “disconnect” or “release tool” is included at the top of the assembly. The release tool enables the operator to selectively disconnect from most of the tools, but there may be a portion of the tool assembly still attached to the end of the wireline that will have to be pulled out of the hole with the wireline. The disconnect may not achieve its intended purpose if the stuck point is above the tool assembly.

(12) Therefore, a typical wireline release additionally utilizes a cable head, which serves as an intentionally added mechanical weak point. The cable head is designed to break upon a predetermined pulling force on the wireline, which is less than the breaking strength of the wireline.

(13) Major problems occur if the cable head and tools are stuck in the well and the wireline breaks upon pulling on the wireline with too much tension. Breaking the wireline and dropping the wireline and tools in the well greatly complicates the fishing operation to retrieve the tools. A contingency means for retrieving stuck wireline is to deploy a device, known as a “cutter”, to cut the wireline at the deepest possible location. It must be attached to the wireline at surface and blindly fall or be intentionally pumped down to a point where the tool assembly is stuck. Wireline cutters that are used will sever the line when it hits either the top of the stuck tool or could hit an obstruction on path which would likely cut the line higher than desired.

(14) Various other apparatus and methods have been provided for releasing the wireline from the cable head and tools. One common method of releasing the wireline includes the use of a spring set at a particular tension. Once the force on the spring is exceeded, the wireline is released. This release still requires that the amount of load required to release the wireline be predetermined prior to lowering the cable head into the well. If the spring tension is exceeded, there can be a premature release of the cable head.

(15) Another release apparatus relies primarily on shear pins or shear screws. Problems are encountered with shear screws having a low failure point because they are exposed to various cyclical forces that tend to affect their ultimate shear rating. The shear screws are exposed to fluids in the well that, over time, can affect the inherent strength of the shear screws or pins making them susceptible to failure at stresses below their rated failure point. Unexpected release can significantly delay operations, causing additional operating expense. An unexpected release can also result in the loss of downhole tools, and in extreme cases can cause severe damage to the wellbore requiring substantial time and money to repair.

(16) Furthermore, most wireline release tools are designed to be installed somewhere in the middle of the tool string below the cable connection point. This means that the wireline may have to pull a portion of the rope socket still attached to the wireline or full outer diameter (OD) tools like casing collar locators, cable head connector and weight bars out of hole in order to be retrieved. It is very common that the obstruction that is causing the tools and wireline to be stuck is above all the tools

at the transition point from wireline OD to tool OD that will cause the wireline to remain stuck.

(17) It is, therefore, desirable to provide a wireline disconnect that ensures the wireline will break where desired and allow for removal of components. As will be described in greater detail below, the invention essentially comprises a selective cut wireline release head that includes a primary device that connects the wireline to the tool string and a device to disconnect the wireline from the tool if the tool has become unintentionally stuck while performing normal operations. The means for connecting the two pieces together may be a set of cups and cones, known as a rope socket cartridge, and may have the ability to transmit electrical current through the entire tool assembly and lock into position so not to swivel in reference to the wireline. A cutting device may be the primary means to release the wireline from the attached tool assembly. The cutting mechanism may be driven with surface controlled incendiary charge that burns within the cable head. When the operation is complete, only the wireline will come out of the hole and the tool string will have to be retrieved with fishing or other tools. At the end of the operation, the pressure from the incendiary will be released automatically prior to retrieving tools.

(18) Referring now to the drawings, and specifically FIG. 1, an exemplary wireline tool string system **1** includes a selective cut wireline release head system **10** according to embodiments of the invention, together with a casing collar logging tool **11**, a perforating gun system **12**, a frac plug setting tool **13**, and a frac plug **14**. Additional, or fewer components, may be included within the tool string system **1**, as is known to those of skill in the art.

(19) While the system **1**, and more particularly the wireline release head system **10**, may be generally used in well wireline operations, and specifically hydrocarbon retrieval, the system **10**, or specific components of the wireline release head system **10**, may be utilized for other well applications such as water retrieval, for example.

(20) Moving on to FIG. 2, the wireline release head system **10** is shown within a well **20**, which is drilled into a surface **30** and includes a lower position **40** where hydrocarbons enter the well **20**. It shall be understood that the system **10** may be used on a horizontal and/or vertical well **20**. The system **10** may generally include a housing **50** with a top sub **60** and a bottom sub **80**. The top sub **60** may include a fishing profile **70**, and weight bars may be attached to the system **10** at the top sub fishing profile **70**.

(21) The housing **50** surrounds an electric wireline **90** which may include a top portion **100**, a middle portion **120**, and a bottom portion **110**. The wireline top portion **100** is operably connected to a control unit **130**. The wireline bottom portion **110** connects to the housing **50** at a rope socket **140**, described in greater detail below.

(22) Between the wireline top portion **100** and the bottom portion **110**, a cutter box **170** surrounds at least a section of the wireline middle portion **120**. The cutter box **170** may have a clamshell design as shown in FIG. 2A. Each half **170a** and **170b** of the cutter box **170** may have a cutout that, when the halves **170a** and **170b** are joined together via fasteners **172**, form an opening for receiving the wireline **90**. The diameter of the opening for the wireline **90** may be substantially similar to the diameter of the wireline **90** itself, but just slightly larger to allow the wireline **90** to freely move up and down within the cutter box **170**. The cutter box **170** may have a threadable attachment portion that allows the cutter box to be threadably attached to the housing **50**. Other means of securing the cutter box **170** to or within the housing **50** are also possible and are within the skill of one in the art.

(23) The cutter box **170** includes a first track **173**. The track **173** provides a translation area for a piston assembly. The piston assembly may include a piston rod **205** and a piston wedge **210**. The piston rod **205** and piston wedge **210** may be separate components, or they may be combined into a single component. Regardless, the piston rod **205** and the piston wedge **210** are activated to move within the track **173** via a power charge **200**, described in greater detail below.

(24) The piston rod **205** may be equipped with seals, such as O-rings, that provide an initial seal between the power charge **200** and the piston wedge **210**, thus allowing the piston rod **205** to push

the piston wedge **210** upon detonation as described below. The seals may be further configured to disengage after a period of time so that pressure is not trapped within the system **10**.

(25) The piston wedge **210** has an angled surface **211**. The angled surface **211** may be angled from about 0 to about 60 degrees relative to vertical. In embodiments, the angle of the piston wedge **210** is about 5 to about 45 degrees, about 5 to about 30 degrees, or **10** to about 20 degrees. In embodiments, the angle of the piston wedge **210** is about 10 degrees. The angled surface **211** of the piston wedge **210** is configured to engage with a cutter knife **150**.

(26) The cutter knife **150** is positioned within a second track **160** defined in the cutter box **170**. The track **160** may be generally perpendicular to or at some other desirable angle relative to track **173**. To facilitate unimpeded movement of the cutter knife **150**, an outside edge **151** of the cutter knife **150** may be angled. The angle of the outside edge **151** of the cutter knife **150** may be determined based on the angle of the angled surface **211** of the piston wedge **210**. In other words, the angled surface **211** of the piston wedge **210** may generally correspond to the angle of the cutter knife outside edge **151**. Corresponding angles between the piston wedge angled surface **211** and the cutter knife outside edge **151** helps smooth translation of the cutter knife **150** within the track **160**, as without corresponding edges, upon detonation, the piston wedge **210** is pushed into the cutter knife **150** which may cause a back end of the cutter knife **150** to be pushed upward within the track **160**, rather than the knife **150** simply sliding towards the wireline **90** within the track **160**.

(27) An inside edge **152** of the cutter knife **150** is also angled so as to provide a sharp edge for slicing through the wireline **90**. It has been found that by reducing the angle of the cutter knife inside edge **152**, shear forces between the cutter knife **150** and the wireline **90** are reduced. The angle of the inside edge **152** may be between about 5 and about 30 degrees relative to horizontal. However, the angle of the cutter knife **150** may be any angle that allows the knife **150** to sever the wireline **90** when the knife **150** is activated.

(28) Moving on, the rope socket **140** may be any rope socket **140** now known or later developed, including but not limited to wedge type, spool type, slip type, and clamp type rope sockets. In embodiments, the wireline **90** is secured to the rope socket **140** via a cup and cone bearing that allows the wireline **90** to rotate within the rope socket **140** as is known to those of skill in the art. The rope socket **140** may be packed with grease to reduce friction on the wireline **90**.

(29) The rope socket **140** may be held in place within the housing **50** via one or more lock screws **220**, further illustrated in FIG. 2B. The lock screws **220** may prevent the rope socket **140** from turning due to torque on the system **10**, and may further prevent the rope socket **140** from moving up and down within the housing **50**.

(30) The wireline **90** extends through the rope socket **140** and is operably connected to an electrical feedthrough **112**, for example, via a tear drop coupling **235**. The tear drop coupling **235** may be seated in a bulkhead **230** such that the electrical feedthrough **112** extends into a lower cavity **245** defined in a bulkhead retainer and switch housing **260**. The bulkhead **230** seals off the cavity **245**, and a secondary seal **240** may be positioned around the coupling **235**, thereby providing pressure isolation between the cavity **245** and the rest of the system **10**.

(31) The electrical feedthrough **112** is operably connected to a fire control switch **180**, which receives a signal from the control unit **130** and subsequently provides electrical signal to the remainder of the system **1** as described below, and controls activation of the cutter tool **150**. The fire control switch **180** may preferably, but need not necessarily, be in line with the wireline **90**. The fire control switch **180** includes electrical leads that may extend through an opening in the bulkhead **230**. Where the electrical leads extend through an opening **250** in the bulkhead **230**, a seal may be placed in the opening to maintain the pressure isolation of the cavity **245**. FIG. 2C is a top view taken along line C-C in FIG. 2, showing the coupling **235** and the opening **250** in the bulkhead **230**. In embodiments, the bulkhead **230** is coupled (e.g., threadably) to an interface member. Like the bulkhead **230**, the interface member may be a metal component such that, when the interface member is coupled to the bulkhead **230**, a metal-to-metal seal is formed between the

interface member and the bulkhead **230**. The electrical leads from the fire control switch **180** may extend through the interface member. The interface member may include, for example, O-rings or other sealing mechanisms for further maintaining the pressure isolation of the cavity when the interface member is coupled to the bulkhead **230**.

(32) However they leave the cavity **245**, the electrical leads are electrically joined to a fire detonator **190**, which is generally housed in a void formed within the housing **50**. The fire detonator **190** is configured to ignite a power charge **200** within the housing **50**. In embodiments, the power charge **200** includes a tube of propellant, such as black powder. The power charge **200** is located beneath the piston rod **205** such that, when the charge **200** is detonated via the fire control switch **180**, the piston rod **205** moves the piston wedge **210** toward the cutter knife **150**, thereby forcing the cutter knife **150** towards the wireline **90** in order to sever the wireline **90**.

(33) The bulkhead retainer and switch housing **260** couples to the bottom sub **80**. The bottom sub **80** may in turn include an electrical coupling mechanism **118**. The electrical coupling mechanism **118** is in electrical communication with the fire control switch **180**, and thus receives signal from the control unit **130** via the fire control switch **180** for providing electrical current to the remainder of the tool assembly below the wireline release head system **10**. The electrical coupling mechanism **118** may be any mechanism now known or later developed. In embodiments, the electrical coupling mechanism **118** is a switch carrier, such as that described in U.S. patent application Ser. No. 17/454,777, which is incorporated by reference herein in its entirety.

(34) Moving on, FIGS. **3**, **3A**, **3B**, and **3C** generally illustrate the position of the cutter knife **150** and the wireline **90** after detonation of the power charge **200**. The control unit **130** selectively sends a signal through the wireline **90** which activates the fire control switch **180** to trigger the fire detonator **190** to detonate the power charge **200**. Detonation of the power charge **200** forces movement of the piston rod **205** and piston wedge **210** in the track **173** towards the cutter knife **150**, as shown by the arrow.

(35) The angled surface **211** of the piston wedge **210** contacts the outside edge **151** of the cutter knife **150**, forcing the cutter knife **150** towards the center of the cutter box **170**, as shown in FIG. **3A**. The force of the piston wedge **210** against the cutter knife **150** causes the cutting edge **152** of the cutter knife **150** to sever the wireline **90**. The end of the wireline **90** may be pushed into free space within the track **160**, as shown by the arrows in FIG. **3**. The remaining freed wireline **90** may be subsequently pulled from the well **20** towards the surface **30**. Fishing operations for the remainder of the tool string can then be commenced.

(36) It shall be understood that the selective cut wireline release head **10** may be placed anywhere along the wireline **90**. If the release head **10** is placed between tools on the wireline **90**, the tools above the release head **10** may be retrieved with the remaining wireline **90** when the wireline **90** is severed as described above. Fishing operations may then be commenced to recover tools that remain in the well **20**.

(37) Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of the disclosure. Embodiments of the invention have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the disclosure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need to be carried out in the specific order described unless specified.

Claims

1. A wireline release head system, comprising: a housing comprising a top sub and a bottom sub; a cutter box coupled to the housing between the top sub and the bottom sub, the cutter box comprising a piston assembly operable to move within a first track, and a cutter knife operable to move within a second track; wherein the second track is angled relative to the first track, wherein at least a portion of the housing surrounds an outer diameter of the cutter box, wherein the cutter box has a clamshell design comprising a first portion and a second portion, the first and second portions together defining a wireline receiving area; a fire control switch in communication with a fire detonator, the fire detonator being selectively activatable to detonate a power charge; and a wireline extending through the cutter box and being in electrical communication with a control unit and the fire control switch; wherein an electrical signal selectively sent through the wireline from the control unit to the fire control switch detonates the power charge, the detonation of the power charge causing the piston assembly to move within the first track and contact the cutter knife, thereby causing the cutter knife to move within the second track, wherein movement of the cutter knife within the second track severs the wireline.
2. The wireline release head system of claim 1, wherein a diameter of the wireline receiving area is substantially similar to a diameter of the wireline.
3. The wireline release head system of claim 1, wherein the piston assembly comprises a piston rod and a piston wedge.
4. The wireline release head system of claim 3, wherein the piston wedge comprises an angled surface.
5. The wireline release head system of claim 1, wherein the piston assembly comprises an angled surface.
6. The wireline release head system of claim 5, wherein: the cutter knife comprises an angled outside edge; and an angle of the angled outside edge of the cutter knife generally corresponds to an angle of the piston assembly angled surface.
7. The wireline release head system of claim 6, wherein the angle of the cutter knife angled outside edge and the piston assembly angled surface is about 5 to about 30 degrees.
8. The wireline release head system of claim 1, further comprising a bulkhead, wherein: the fire control switch is positioned within a cavity in the housing; and the bulkhead is positioned atop the cavity to isolate pressure within the cavity.
9. The wireline release head system of claim 8, further comprising a rope socket, wherein the rope socket receives an end of the wireline and is electrically coupled to an electrical feedthrough, the electrical feedthrough extending into the cavity and being electrically coupled to the fire control switch.
10. The wireline release head system of claim 9, wherein the electrical feedthrough is coupled to the rope socket via a tear drop coupling.
11. The wireline release head system of claim 1, further comprising a rope socket, wherein the rope socket receives an end of the wireline and is electrically coupled to an electrical feedthrough, the electrical feedthrough being electrically coupled to the fire control switch.
12. The wireline release head system of claim 1, wherein the fire control switch is in electrical communication with an electrical coupling mechanism, the electrical coupling mechanism being configured to provide electrical current from the control unit to a tool coupled to the wireline release head.
13. The wireline release head system of claim 1, wherein the second track is generally perpendicular to the first track.
14. A wireline release head, comprising: a cutter box coupled to a housing between a top sub and a bottom sub, wherein at least a portion of the housing surrounds an outer diameter of the cutter box, wherein the cutter box has a clamshell design comprising a first portion and a second portion, the first and second portions together defining a wireline receiving area, the cutter box comprising: a

piston assembly operable to move within a first track defined in the cutter box; a cutter knife operable to move within a second track defined in the cutter box; and a wireline receiving area for receiving at least a portion of a wireline; wherein: the second track is angled relative to the first track; and activation of movement of the piston assembly within the first track causes the piston assembly to contact the cutter knife thereby causing the cutter knife to move within the second track, wherein movement of the cutter knife within the second track severs the wireline.

15. The wireline release head of claim 14, wherein the piston assembly comprises an angled surface.

16. The wireline release head of claim 15, wherein: the cutter knife comprises an angled outside edge; and an angle of the angled outside edge of the cutter knife generally corresponds to an angle of the piston assembly angled surface.

17. The wireline release head of claim 14, wherein: a diameter of the wireline receiving area is substantially similar to a diameter of the wireline.

18. The wireline release head of claim 14, wherein the second track is generally perpendicular to the first track.

19. A method of severing a wireline, comprising: providing a wireline release head, comprising: a cutter box coupled to a housing between a top sub and a bottom sub, the cutter box comprising a piston assembly operable to move within a first track, and a cutter knife operable to move within a second track, wherein at least a portion of the housing surrounds an outer diameter of the cutter box, wherein the cutter box has a clamshell design comprising a first portion and a second portion, the first and second portions together defining a wireline receiving area; a fire control switch for selectively detonating a power charge; and a wireline extending through the cutter box and being in electrical communication with a control unit and the fire control switch; and selectively activating the control unit to send an electrical signal through the wireline to the fire control switch; wherein the fire control switch detonates the power charge upon receipt of the electrical signal, the detonation of the power charge causing the piston assembly to move within the first track and contact the cutter knife, thereby causing the cutter knife to move within the second track, the movement of the cutter knife within the second track severing the wireline.

20. The method of claim 19, wherein the fire control switch is in electrical communication with an electrical coupling mechanism, the electrical coupling mechanism being configured to provide electrical current from the control unit to a tool coupled to the wireline release head.
